# Project 6

**Keypad and menu interface**

EELE 465

## Project overview

**Project Goal** Create a physical UI that allows users to navigate a menu system on the LCD using a rotary encoder and enter values using a keypad.

In this project, you'll add physical UI elements to your project 5 system and create a menu system that is displayed on the LCD. The menu system will let users view and change system settings. A 3x4 matrix keypad will be connected to an MSP430FR2310, which will send the keypad output to the main MSP430FR2153 via UART. A rotary encoder will be connected to the MSP430FR2153, which will be used to navigate the menu system. An RGB LED will indicate the system's state.

All previous functionality from project 5 will be maintained *except for the following*:

- The Python GUI and the USB-C-to-UART adapter. You will replace the GUI's UART connection with the MSP430FR2310's UART connection. All settings will be set through the menu system instead of your computer.
- The button that changes the WS2812B LED color; the LED color will be set via the menu system.

Concretely, that means you need the following hardware from the previous projects:

- A reset button.
- A heartbeat LED.
- The LED patterns on an LED bar.
- A button to change the LED pattern.
- A slide potentiometer to change the LED pattern speed.
- The WS2812B LED stick that shows the potentiometer position.
- The LCD.
- A DAC to control the LCD contrast.
- The temperature sensor.
- The $I^2C$ RTC.

## Outcomes

After this project, you will be able to do the following:

- Scan a matrix keypad.
- Use a quadrature rotary encoder for user input.
- Create a menu system on an LCD.
- Communicate between two MCUs using UART.

# Achievement levels

In general, I don't expect all students to complete all requirements. The requirements are roughly divided into "achievement levels" based upon estimated difficulty and how integral they are to the project overall. The following emojis indicate the achievement levels:

- 👌 proficient; between 60%–80%.
- 👍 strong; between 80%–90%.
- 💯 excellent; between 90%–100%.

The achievement levels assigned to each requirement are my subjective estimations. You may choose whichever requirements you want to do to achieve your final grade. I understand these projects are challenging and I have high expectations; grades will be curved accordingly at the end of the semester if needed.

# Project description

Again, the main goal of this project is to create a menu system on the LCD that can be navigated using a rotary encoder and keypad. The keypad will be connected to a separate microcontroller (MSP430FR2310); the MSP430FR2310 (keypad MCU) and the MSP430FR2153 (main MCU) will communicate via UART. Additionally, an RGB LED will indicate the system's state.

Besides the addition of the menu system, this project's functionality is nearly identical to project 5: you will be reading from the RTC, displaying LED patterns, controlling LED pattern speed with a slide potentiometer, reading an analog temperature sensor, and displaying all the associated information on the LCD. The LCD's menu system will allow users to view and change settings, replacing the UART and GUI connection from your main computer.

## Main screen

The main LCD screen, shown in figure 1, will be the same as in project 5.

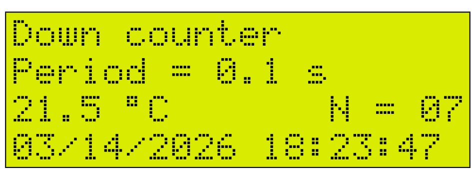


Figure 1: Example of the LCD text layout.

## Menu system

The menu system will allow users to view and change system settings. Each menu screen will have a list of menu items that can be selected. The menu system comprises four item types:

- previous screen
- submenu
- entry

- toggle

Figure 2 shows an abstract example of the menu system and the different item types.

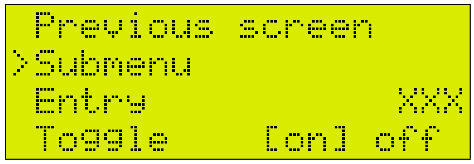


Figure 2: Example of the different menu item types. The currently selected menu item will have an indicator next to it in the leftmost position.

Each menu will have an item at the top that takes users back to the previous screen (either a menu or the main screen). Submenu items will bring up another menu. Entry item types allow users to enter numeric values using the keypad. Toggle item types allow users to toggle between two settings. The currently selected menu item will have an indicator displayed next to the item name.

A menu may have more than four menu items, in which case the displayed menu items must change when the user scrolls to a menu item that is not currently displayed.

Rotating the rotary encoder will navigate through the menu items; a clockwise rotation will select the next menu item (moving down), while a counterclockwise rotation will select the previous menu item (moving up). Pressing the rotary encoder's button will perform the action association with selected menu item, as described in Table 1.

Table 1: Menu item types and their associated actions when pressing the rotary encoder button. "Primary button action" indicates what happens the rotary encoder button is pressed for the first time; most item types only have a primary button action. For the entry item type, "secondary button action" indicates what happens when the rotary encoder button is pressed a second time.

| **Menu item type** | **Primary button action** | **Secondary button action** |
|---|---|---|
| Previous screen | Go to the previous screen. | |
| Submenu | Go to the selected submenu. | |
| Entry | Enter the value-entry state that allows users to enter values with the keypad. | Confirm the entered value and return control to the menu. |
| Toggle | Toggle which value is selected. This makes the unselected setting the new active setting. | |

When on the main the screen, pressing the rotary encoder button will cause the system to enter the main menu.

The following sections describe each menu and their associated menu items. When applicable, the layout of the LCD screen is shown.

### Main menu

Table 2 shows the main menu's items and their associated types and actions.

Table 2: Main menu items and their associated types and actions.

| Menu item | Item type | Action |
|---|---|---|
| Main screen | Previous screen | Go to the main screen. |
| Window size | Entry | Set the temperature sensor's moving average window size. |
| LCD contrast | Entry | Set the LCD's contrast. |
| RTC settings | Submenu | Display the RTC settings menu. |
| WS2812B color | Submenu | Display the WS2812B color settings menu. |
| Cursor | Toggle | Turn the LCD cursor on or off. |
| Cursor blink | Toggle | Turn the LCD cursor's blink function on or off. |

Figures 3, 4, and 5 show examples of how the main menu might look. Note that the main menu has more than four items, so the menu will need to scroll when an item that is off the screen is selected.

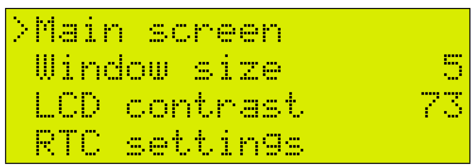


Figure 3: Example of the first four main menu items being displayed. The "main screen" item is currently selected, as indicated by the greater-than symbol.

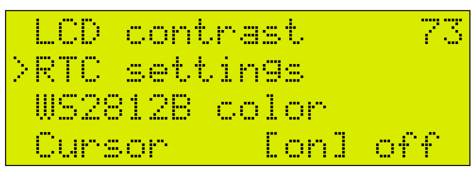


Figure 4: Example of some middle main menu items being displayed—neither the top nor bottom menu items are visible. The "RTC settings" item is currently selected, as indicated by the greater-than symbol.

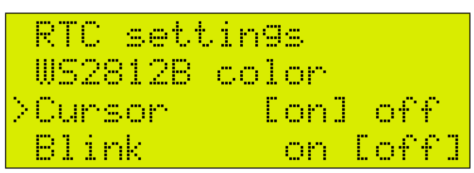


Figure 5: Example of the last four main menu items being displayed. The "cursor" item is currently selected, as indicated by the greater-than symbol.

### RTC settings menu

The RTC settings menu is a submenu that will be accessed from the main menu. This menu will be used to configure the RTC's date and time registers. Table 3 shows the RTC settings menu items, while figures 6 and 7 show examples of what the menus might look like.

Table 3: RTC settings menu items and their associated types and actions.

| Menu item | Item type | Action |
|---|---|---|
| Main menu | Previous screen | Go to the main menu. |
| Year | Entry | Set the RTC's year. |
| Month | Entry | Set the RTC's month. |
| Date | Entry | Set the RTC's date. |
| Hour | Entry | Set the RTC's hour. |
| Minute | Entry | Set the RTC's minute. |
| Second | Entry | Set the RTC's second. |
| AM/PM (optional) | Toggle | If you're using AM/PM mode, set AM or PM. |

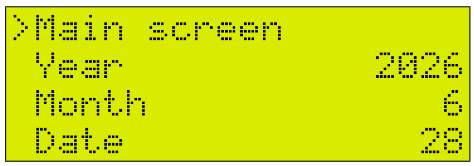


Figure 6: Example of the first four RTC settings menu items being displayed.

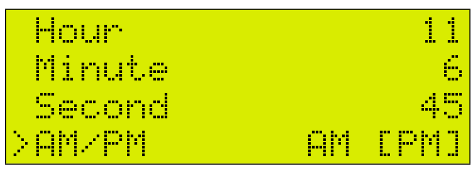


Figure 7: Example of the last four RTC settings menu items being displayed.

### WS2812B color menu

The WS2812B color menu is a submenu that is accessed from the main menu. This menu will be used to set the RGB value of the WS2812B LEDs (all LEDs will be the same color...😞). Table 4 describes the menu items and figure 8 shows an example LCD layout.

Table 4: RTC settings menu items and their associated types and actions.

| Menu item | Item type | Action |
|---|---|---|
| Main menu | Previous screen | Go to the main menu. |
| Red | Entry | Set the red value between 0 and 255. |
| Green | Entry | Set the green value between 0 and 255. |
| Blue | Entry | Set the blue value between 0 and 255. |


Figure 8: Example of the WS2812B color menu

### Menu hierarchy summary

The following list shows a compact view of the entire menu structure. The icons indicate each item's type. You don't need to have the menu items in this exact order.

- ← Main screen
- ⌨ Window size
- ⌨ LCD contrast
- → RTC settings
  - ← Main menu
  - ⌨ Year
  - ⌨ Month
  - ⌨ Date
  - ⌨ Hour
  - ⌨ Minute
  - ⌨ Second
  - ◐ AM/PM (optional)
- → WS2812B color
  - ← Main menu
  - ⌨ Red
  - ⌨ Green
  - ⌨ Blue
- ◐ Cursor
- ◐ Blink

# Project requirements

## Project 5 functionality

**Requirement 1** All hardware peripherals from project 5 are functional. 6 pts 👌

## Keypad MCU (MSP430FR2310)

The MSP430FR2310 needs to be soldered onto a breakout board. This MCU will be used to scan the keypad, hence it will be referred to as the "keypad MCU".

**Requirement 2** The keypad MCU must have a heartbeat LED. 1 pt 👌

**Requirement 3** The keypad MCU must accept input from a 3x4 matrix keypad. 20 pts 👌

**Requirement 4** The keypad MCU must communicate keypad inputs to the main MCU via UART. 10 pts 👌

**Requirement 5** The keypad MCU must have an LED that lights up when a keypad button is pressed. 1 pt 👌

**Specification 5.1** The LED must remain lit long enough for users to see it.

## Rotary encoder

The rotary encoder will be connected to the main MCU (MSP430FR2153).

**Requirement 6** The main MCU must capture clockwise and counterclockwise rotary encoder rotations. 16 pts 👌

**Requirement 7** The main MCU must capture rotary encoder button presses. 3 pts 👌

## Main screen

**Requirement 8** The main screen display is the same as specified in project 5. 3 pts 👌

## Menu system

**Requirement 9** Pressing the rotary encoder button on the main screen enters the main menu. 6 pts 👌

**Requirement 10** The rotary encoder is used to scroll through the menu items. 6 pts 👌

**Specification 10.1** Clockwise rotation scrolls down.

**Specification 10.2** Counterclockwise rotation scrolls up.

**Specification 10.3** The top and bottom menu items can't be scrolled past.

**Requirement 11** Pressing the rotary encoder button on a menu item performs the action associated with the menu item.

**Requirement 11.1** Previous screen: the current menu is exited and the previous screen is displayed. 4 pts 👌

**Specification 11.1.1** If the previous screen is a menu, the previous menu's state doesn't have to be restored.

**Requirement 11.2** Submenu: the submenu is entered and displayed. 1 pt 👍

**Requirement 11.3** Toggle: the active value is changed. 4 pts 👌

**Requirement 11.4** Entry: the first button press lets users enter a value with the keypad, and the second button press confirms the value. 4 pts 👌

**Requirement 12** The currently selected menu item has an indicator to the left of the menu item's name (see the LCD figures for an example). 2 pts 👌

**Requirement 13** The currently selected menu item indicator is a custom character. 1 pt 💯

**Requirement 14** Toggle menu items must indicate which option is active. 2 pts 👌

**Requirement 15** Toggle menu items use a custom character to indicate which option is active. 0.5 pts 💯

**Requirement 16** All "previous screen" menu items display a custom character that visually indicates that the user is moving back to the previous screen. 0.5 pts 💯

### Main menu

**Requirement 17** Selecting the "Main screen" menu item must return to the main screen. 2 pts 👌

**Requirement 18** The user must be able to set the temperature sensor's moving average window size. 2 pts 👌

**Requirement 19** The user must be able to set the LCD contrast. 4 pts 👍

**Requirement 20** The user must be able to enter the RTC settings submenu. 0.5 pts 👍

**Requirement 21** The user must be able to enter the WS2812B color submenu. 0.5 pts 👍

**Requirement 22** The user must be able to turn the LCD's cursor on and off. 2 pts 👌

**Requirement 23** The user must be able to turn the LCD cursor's blinking on and off. 2 pts 👌

### RTC settings main

**Requirement 24** The user must be able to return to the main menu.

**Requirement 25** The user must be able to set the RTC's year. 0.5 pts 👍

**Requirement 26** The user must be able to set the RTC's month. 0.5 pts 👍

**Requirement 27** The user must be able to set the RTC's date. 0.5 pts 👍

**Requirement 28** The user must be able to set the RTC's hour value. 0.5 pts 👍

- **Requirement 28.1** If the RTC is set to 12-hour mode, the user must be able to configure AM or PM.

**Requirement 29** The user must be able to set the RTC's minute value. 0.5 pts 👍

**Requirement 30** The user must be able to set the RTC's second value. 0.5 pts 👍

### WS2812B color menu

**Requirement 31** The user must be able to return to the main menu.

**Requirement 32** The user must be able to set the red, green, and blue values for the WS2812B LEDs. 3 pts 👍

- **Specification 32.1** Each LED will be the same color.
- **Specification 32.2** RGB values must range between 0–255.

## RGB state indicator LED

An RGB LED will indicate the system's state as defined in the following requirements.

**Requirement 33** The RGB LED must show a unique color for each LED pattern when the system is on the main screen. 4 pts 💯

**Requirement 34** The RGB LED must show a unique color when the system is in the menu system. 2 pts 💯

**Requirement 35** The RGB LED must show a unique color when a value is being entered in the menu system—i.e., when an "entry" menu item is active and the system is accepting keypad input). 2 pts 💯

> ℹ The RGB state-indicator LED isn't difficult per se, but implementing it will make you run out of pins and need to use the $I^2C$ port expander—hence the 💯 achievement level.

## Settings storage

MSP430FR MCUs use ferroelectric RAM (FRAM) for their *non-volatile* storage instead of typical flash memory. FRAM is interesting because it has fast access times like traditional RAM but is non-volatile; these properties allow FRAM to be used for both program storage (which is usually stored in flash) and variable storage (which is usually stored in DRAM or SRAM). Variables stored in FRAM are persistent across power cycles—that is, they retain their value. FRAM lets us have persistent variables without needing an external EEPROM.

The MSP430 compiler has a `#pragma PERSISTENT` directive that lets us put variables in FRAM instead of SRAM. We will use this feature to make our settings persistent. See the resources in Canvas under `Resources/FRAM` for more information on FRAM; see the MSP430 C/C++ compiler manual for more information about the available `#pragma` directives.

**Requirement 36** The following settings are persistent and retain their value after a reboot or power cycle: 2 pts 💯

- window size
- LCD contrast
- WS2812B color
- cursor state
- cursor blink state

> ℹ Variables in FRAM get initialized every time the debugger flashes the code, so your variables will be reset every time you debug or flash the code.

# Demo

1. Introduction
    - Introduce the project at a high level. Don't assume the instructor or TA knows about the project. Focus on the goal of the project.
    - Use a software architecture diagram to discuss how the software was *designed*.
    - Use a high-level flowcharts to help describe the system's functionality.
    - Use a circuit diagram to explain how the circuit was constructed.
2. Demonstration
    - Demonstrate the full system working. You may choose the order in which you would like to present functionality.
    - Be sure to demonstrate that every specification was met.
    - The partner that was primarily responsible for a subsystem/requirement/specification should present that functionality.